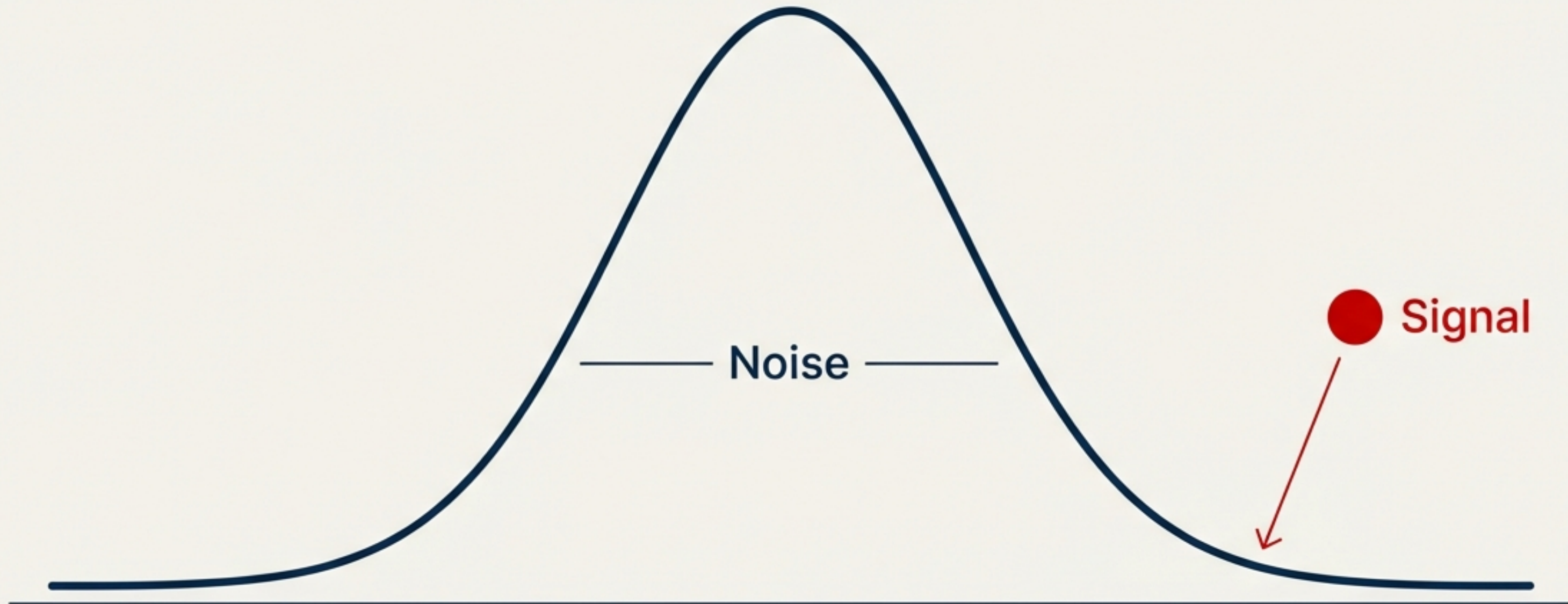


Z-TESTS: THE SCIENCE OF SIGNIFICANCE

Distinguishing Signal from Noise in Data Decisions



BASED ON ONE-SAMPLE AND TWO-SAMPLE STATISTICAL METHODS

Is the difference real, or is it just random chance?

A Z-test is a statistical method used to validate claims. It measures how many standard deviations a sample is away from the population mean.

Low Z-Score:
Likely Random Chance.

Low Z-Score
Likely Random Chance

High Z-Score:
Likely Real Impact.



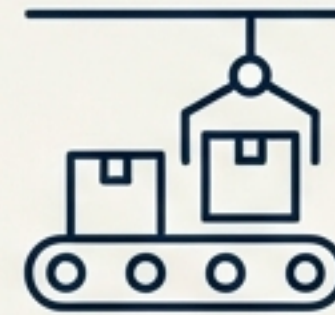
How do we know if a result is statistically significant?

Turning Data into Verdicts Across Industries



Medical Trials

Does this new drug reduce symptoms better than the current standard?



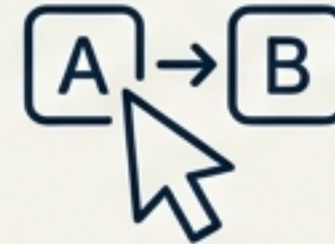
Manufacturing Quality

Is the bottling machine underfilling boxes against the 500ml standard?



Education

Did the new training programme actually improve student scores?



Website A/B Testing

Does Version A lead to significantly more clicks than Version B?

The Z-Test replaces guesses with statistically valid evidence.

The Tool: One-Sample Z-Test

Checks whether the mean of a single sample is significantly different from a known population mean.

The diagram illustrates the One-Sample Z-Test formula, $Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$, with labels for its components:

- Sample Mean** (The evidence we found) points to $\bar{X}$.
- Population Mean** (The standard we test against) points to μ .
- Population Standard Deviation** points to σ .
- Sample Size** points to n .

The formula is displayed as:

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

Investigation 1: The Bottling Machine

The Scenario:

A machine is expected to fill **500 ml** (μ). We suspect it is underfilling.

The Evidence:

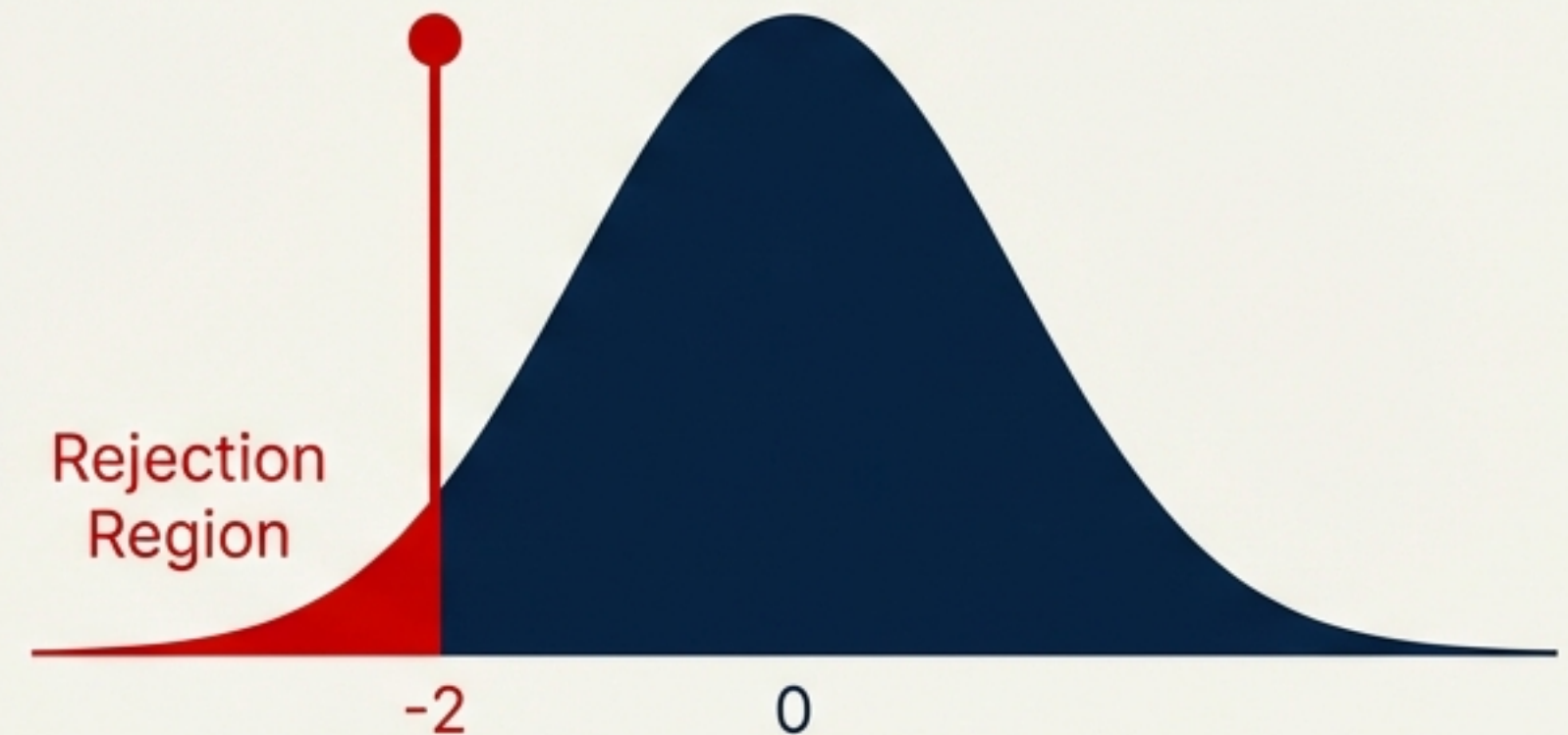
- Sample Size (n): 36 bottles
- Sample Average ($\bar{X}$): 492 ml
- Variation (σ): 24 ml

Calculation:

$$Z = (492 - 500) / (24 / \sqrt{36})$$

$$Z = -8 / 4$$

Result: $Z = -2$



Investigation 2: Validating the School's Claim

The Scenario:

A school claims the average IQ is **100** (μ).
A researcher suspects this is incorrect
(could be higher or lower).

Calculation:

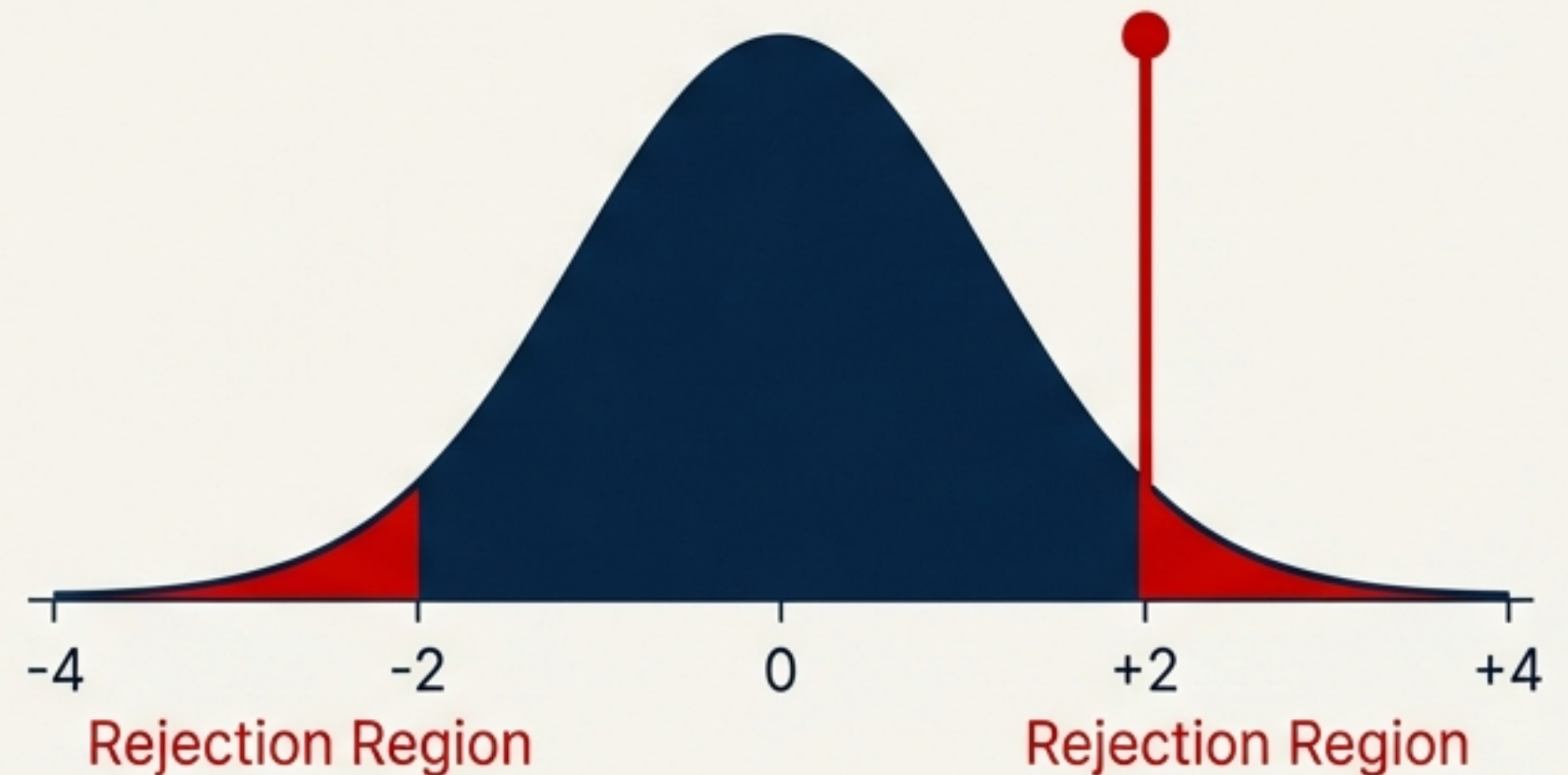
$$Z = (104 - 100) / (14 / \sqrt{49})$$

$$Z = 4 / 2$$

Result: $Z = 2$

The Evidence:

- Sample Size (n): 49 students
- Sample Average ($\bar{X}$): 104
- Variation (σ): 14



Translating the Score: Reading the P-Value

A Z-score is just a coordinate. The P-value is the probability.

Look up $Z = 2.00$

Z	0.00
1.0	0.9837
1.5	0.9729
2.0	0.9772
2.5	0.9772

Standard cumulative area from the left.



Find the Tail Area

$$1 - 0.9772 = \mathbf{0.0228}$$

Area to the right of $Z = 2.00$.



Two-Tailed Adjustment

$$2 \times 0.0228 = \mathbf{0.0456}$$

For a two-tailed test, checking both extremes.



Verdict:

Since $P(0.0456) < 0.05$, we **Reject the Null Hypothesis**.

The difference is significant.

The Comparison: Two-Sample Z-Test

Used when comparing the means of two independent populations ($n > 30$).

$$Z = \frac{\overbrace{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}^{\text{Difference in Means}}}{\underbrace{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}_{\text{Combined Variance}}}$$

Comparison 1: School A vs. School B

School A

Mean=75

$\sigma = 10$

$n = 100$

School B

Mean=72

$\sigma = 12$

$n = 120$

Calculation Steps:

Step 1:

Difference in means: $75 - 72 = 3$

Step 2:

Combined Variance: $\sqrt{(100/100) + (144/120)} = 1.483$

Step 3:

Z-Score: $\frac{3}{1.483} \approx 2.022$

P-value = 0.0436. Result is Significant (< 0.05).

Comparison 2: Measuring Training Impact

Does the Trained Group (A) score higher than the Untrained Group (B)?

Group A (Trained)

Mean=82

$\sigma=8$

$n=64$

Group B (Untrained)

Mean=78

$\sigma=6$

$n=49$

$$\frac{82 - 78 = 4}{\approx 1.3133} = 3.045$$

$Z =$

Conclusion: P-value (0.0012) < 0.05. The training is a verified success.

Automating the Analysis with Python

```
import numpy as np
from scipy.stats import norm

# Input Data
X1, sigma1, n1 = 75, 10, 100
X2, sigma2, n2 = 72, 12, 120

# Calculate Z-Score
z = (X1 - X2) / np.sqrt((sigma1**2)/n1 + (sigma2**2)/n2)
# Output: Z-Score: 2.022

# Calculate P-Value (Two-tailed)
p = 2 * (1 - norm.cdf(abs(z)))
# Output: p-value: 0.0436
```

Logic: If $p < 0.05$: Reject H_0 (Significant Difference).

The Decision Matrix

The P-Value Rule

$P < 0.05$ → Unlikely to be random.
Reject Null Hypothesis.
(Significant)

$P > 0.05$ → Could be chance. **Fail to**
Reject Null Hypothesis.
(Not Significant)

When to use Z-Test?

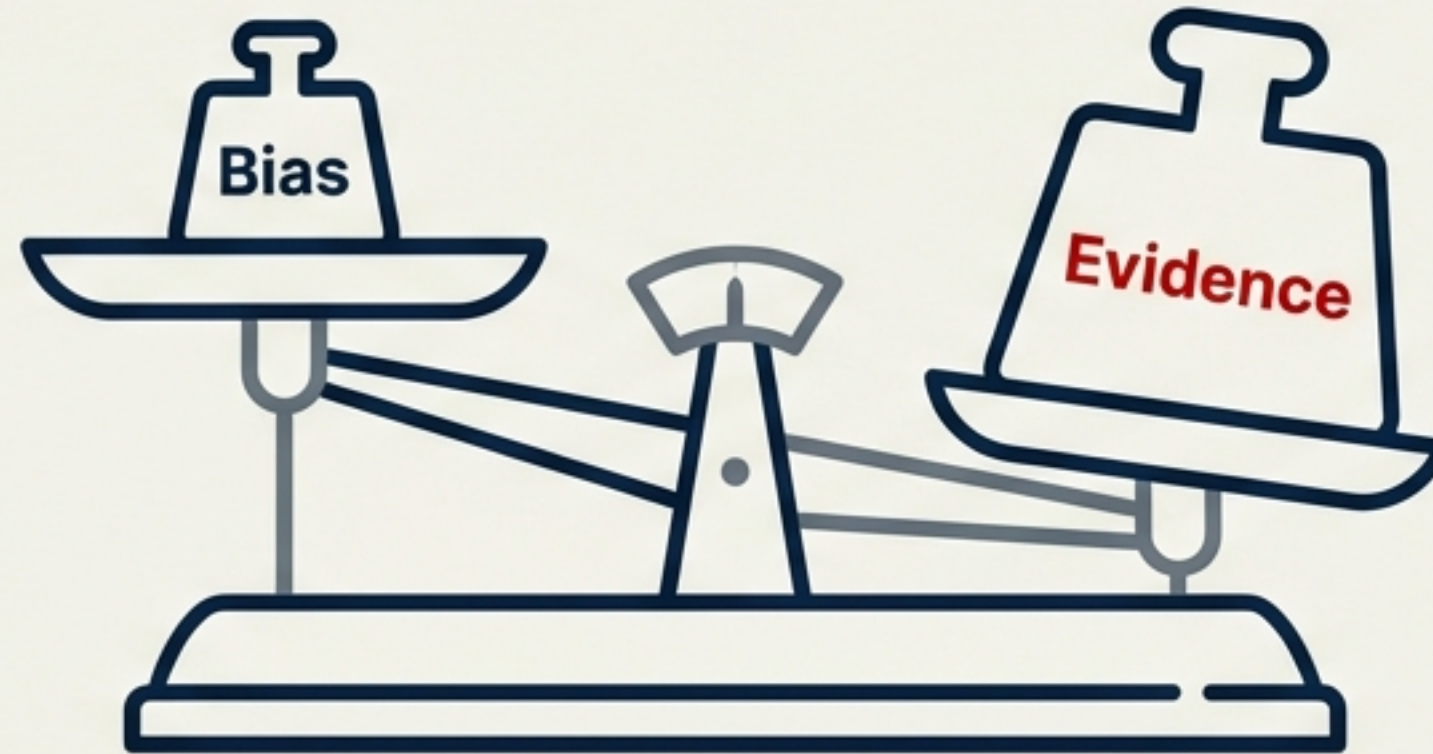
1. Population **variance** is **known**.
2. Sample size is **large** ($n > 30$).
3. Samples are **independent**.

Test Types

One-Tail: Testing direction (Better/Worse).

Two-Tail: Testing for *any* difference.

Certainty in an Uncertain World



The Z-Test does not give us 'truth' in the absolute sense. It gives us a calculated probability of error. It allows us to move from 'I think' to 'The evidence suggests'.